

Choose the correct answers.

1. A practical example is the assembly of pistons with cylinder bores. Let the bore size be 50 mm and the clearance required for the assembly is 0.12 mm on the diameter. Let the tolerance on bore and the piston each = 0.04 mm. Find
 - i. The base size of the piston
 - ii. The Upper and Lower deviation
 - a. $i = 50$, $ii = 49.88$
 - b. $i = 49.88$, $ii = 0.02$
 - c. $i = 0.02$, $ii = 50$
 - d. $i = 0.02$, $ii = 49.88$
2. Two shafts A and B have their diameters specified as 100 ± 0.1 mm and 0.1 ± 0.0001 mm respectively. Which of the following statements is/are true?
 - a. Tolerance in the dimension is greater in shaft A
 - b. The relative error in the dimension is greater in shaft A
 - c. Tolerance in the dimension is greater in shaft B
 - d. The relative error in the dimension is the same for shaft A and shaft B
3. A shaft has a dimension, $35\phi_{-0.025}^{-0.009}$. The respective values of fundamental deviation and tolerance are
 - a. -0.025 , ± 0.008
 - b. -0.025 , 0.016
 - c. -0.009 , ± 0.008
 - d. -0.009 , 0.016
4. A hole is specified as 40 ± 0.050 mm. The mating shaft has a clearance fit with a minimum clearance of 0.01 mm. The tolerance on the shaft is 0.04 mm. The maximum clearance in mm between the hole and the shaft is:
 - a. 0.04
 - b. 0.05
 - c. 0.10
 - d. 0.15
5. Interference fit joints are provided for:
 - a. Assembling bush bearing in housing
 - b. Mounting heavy duty gears on shafts
 - c. Mounting pulley on shafts
 - d. Assembly of flywheels on shafts
6. In order to have interference fit, it is essential that the lower limit of the shaft should be:
 - a. Greater than the upper limit of the hole
 - b. Lesser than the upper limit of the hole
 - c. Greater than the lower limit of the hole
 - d. Lesser than the lower limit of the hole
7. In an interchangeable assembly, shafts of size $25_{-0.01}^{+0.04}$ mm mate with holes of size $25_{-0.02}^{+0.02}$ mm. The maximum possible clearance in the assembly will be:
 - a. 10 microns
 - b. 20 microns
 - c. 30 microns
 - d. 60 microns
8. Dimension of the hole is $50_{-0.00}^{+0.02}$ mm and shaft are $50_{+0.00}^{+0.02}$ mm. The minimum clearance is:
 - a. 0.02 mm
 - b. 0.00 mm
 - c. -0.02 mm
 - d. 0.01 mm
9. In an interchangeable assembly, shafts of size $25_{-0.01}^{+0.04}$ mm mate with holes of size $25_{-0.02}^{+0.03}$ mm. The maximum interference (in microns) in the assembly is:

- a. 40
b. 30
- c. 20
d. 10
10. Which one of the following is a clearance fit?
- a. $\phi H50^{+0.015}_{+0.005} h\phi 50^{-0.010}_{-0.000}$
b. $\phi H50^{+0.01}_{+0.00} h\phi 50^{+0.025}_{+0.015}$
c. $\phi H50^{+0.015}_{+0.00} h\phi 50^{+0.025}_{+0.005}$
d. $\phi H50^{-0.01}_{-0.00} h\phi 50^{+0.030}_{+0.005}$
11. A shaft and hole pair is designated as 50H7d8. This assembly constitutes:
- a. Interference fit
b. Transition fit
c. Clearance fit
d. None of the above
12. Which of the following is an interference fit?
- a. Push fit
b. Running fit
c. Sliding fit
d. Shrink fit
13. Consider the following fits:
- I.C. engine cylinder and piston
 - Ball bearing outer race and housing
 - Ball bearing inner race and shaft
- Which of the above fits are based on the interference system?
- a. i and ii
b. ii and iii
c. i and iii
d. i, ii and iii
14. Allowance in limits and fits refers to:
- Maximum clearance between shaft and hole
 - Minimum clearance between shaft and hole
 - Difference between maximum and minimum size of hole
 - Difference between maximum and minimum size of shaft
15. Clearance in a fit is the difference between:
- Maximum hole size and minimum shaft size
 - Minimum hole size and maximum shaft size
 - Maximum hole size and maximum shaft size
 - Minimum hole size and minimum shaft size
16. Consider the following statements: A nomenclature $\phi 50 H8/p8$ denotes that:
- Hole diameter is 50 mm.
 - It is a shaft base system.
 - 8 indicates fundamental deviation. Which of the statements given above is/are incorrect?
- a. i, ii and iii
b. i and ii only
c. i and iii only
d. i only
17. The dimensional limits on a shaft of 25h7 are:
- 25.000, 25.021 mm
b. 25.000, 24.979 mm
c. 25.000, 25.007 mm
d. 25.000, 24.993 mm
18. Following data are given for calculating limits of dimensions and tolerances for a hole: Tolerance unit i (in μm) = $0.45 \sqrt[3]{D} + 0.001D$. The unit of D is mm. Diameter step is 18-30

mm. If the fundamental deviation for H hole is zero and $IT8 = 25 \mu$, the maximum and minimum limits of dimension for a 25 mm H8 hole (in mm) are:

- a. 24.984, 24.967
 - b. 25.033, 25.000
 - c. 25.017, 24.984
 - d. 25.000, 24.967
19. A fit is specified as 25H8/e8. The tolerance value for a nominal diameter of 25 mm in IT8 is 33 microns and fundamental deviation for the shaft is -40 microns. The maximum clearance of the fit in microns is:
- a. -7
 - b. 7
 - c. 73
 - d. 106

Calculate the following questions. Show the necessary steps.

20. Determine the Limits on a shaft and a pulley of 40 mm nominal diameter. The tolerance on shaft is h6 and on the pulley is H7. Calculate:
- a. The tolerance on the shaft.
 - b. The tolerance on the pulley.
 - c. The maximum clearance.
 - d. The minimum clearance.
21. Determine the Limits on a shaft and a pulley of 45 mm nominal diameter. The tolerance on the shaft is g6 and on the pulley is H7. Calculate:
- a. The tolerance on the shaft.
 - b. The tolerance on the pulley.
 - c. The maximum clearance.
 - d. The minimum clearance.